Student Name: Zeyad Sherif **Student ID: 202201220** **GitHub_Repo**

DSAI 940 - Assignment 1 Report Video_Demo → Drive

1. Model Architectures & Implementation

This assignment investigates representation learning by training separate Autoencoder (AE) and Variational Autoencoder (VAE) models for six distinct anatomical regions (AbdomenCT, BreastMRI, CXR, ChestCT, Hand, HeadCT). Developing distinct models prevents latent space entanglement across highly varied medical textures and isolates the generative and reconstruction features of each class.

Denoising Autoencoder (DAE):

The standard Autoencoder was converted into a Denoising Autoencoder in order to analyse robustness. It uses a Convolutional Encoder made of three Conv2D layers (32, 64, and 128 filters each) with strides of 2 to downsample the input (64 x 3) to a flattened 16-dimensional deterministic latent-layer vector.

Variational Autoencoder (VAE):

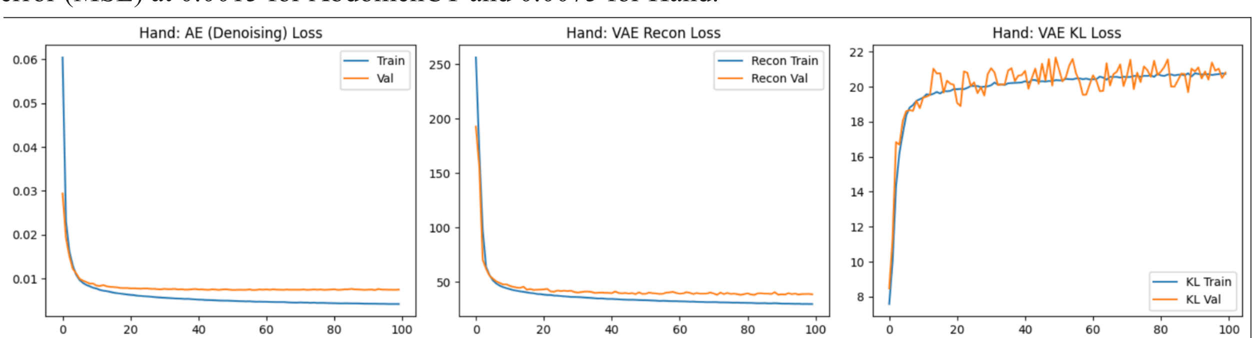
The VAE uses the same Convolutional Encoder but has a Probabilistic Bottleneck instead of a discrete vector like the AE does. The Encoder outputs a mean and a log-variance rather than a deterministic vector. By applying the reparameterization trick, the VAE samples from the probabilistic distribution of the values provided by the Encoder.

2. Key Differences & Training Observations

The AE is a deterministic model hence restoration is really effective at the pixel level but creates gaps in its latent space which greatly restricts generative potential. The VAE creates a latent space that is structured and continuous although it sacrifices some of the sharpness of raw reconstructions to achieve such a latent space that can be used to sample entirely new data.

Training Dynamics:

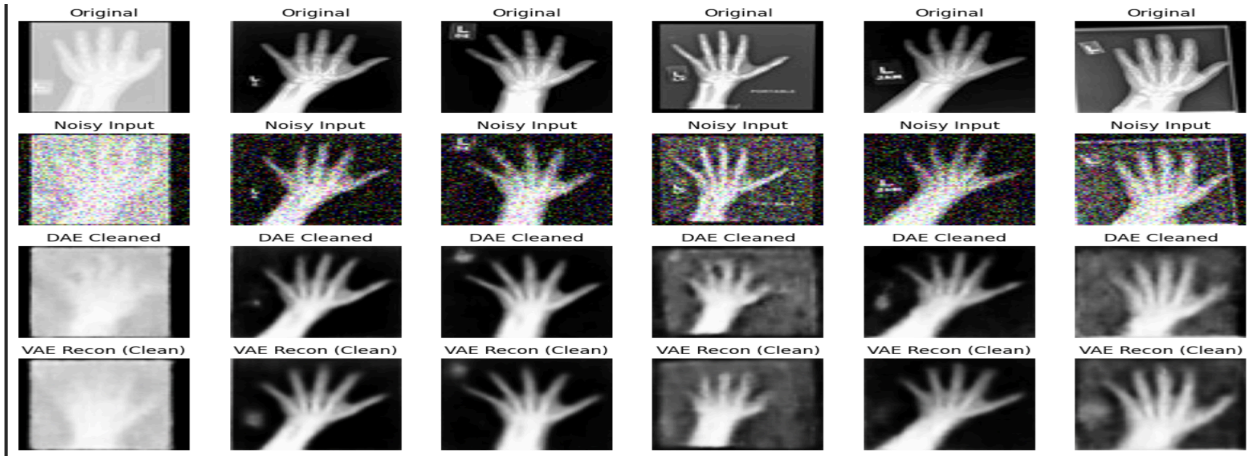
After training for 100 epochs, the Denoising AE had consistently achieved low validation mean squared error (MSE) at 0.0015 for AbdomenCT and 0.0075 for Hand.



3. Reconstruction & Denoising Robustness

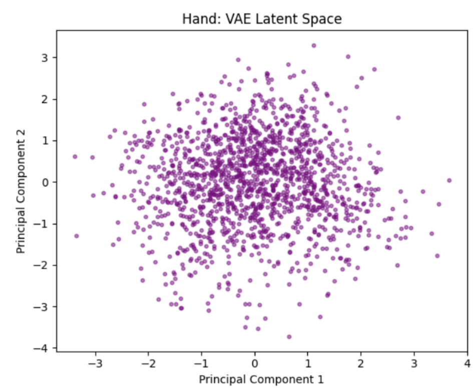
The DAE performed very well in its intended role. Even though there was a high amount of 20%

Gaussian noise added, the DAE detected all of the aspects of the underlying anatomy in the data and was able to generate (output) new, clean, and smooth representations of the original data.



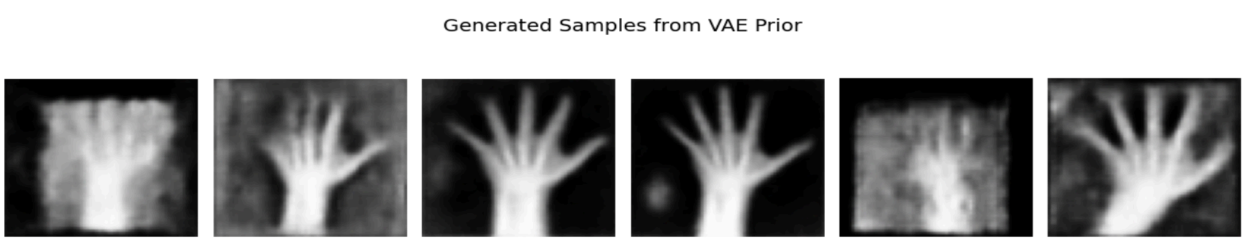
The Variational Autoencoder (VAE) reconstructed images were slightly blurred, an established effect of both the Mean Square Error (MSE) loss function and the variability with which samples are broadcasted into a Latent Space.

4. Latent Space Behavior & Sample Generation



The true advantage of the VAE lies in its latent representations. Principal Component Analysis (PCA) was used to project the 16-dimensional mu vectors into 2D space.

In addition to the shift from the original prior distributions of the Latent Variables to a standard normal prior distribution, the result is that the Latent Space will be filled continuously with no area being dead. Thus, points in the Latent Space may be easily interpolated between or generated entirely through random sampling.



Conclusion & Insights:

These findings successfully confirm the theoretical differences in function between standard and variational autoencoders (VAEs). The Denoising Autoencoder (DAE) serves as a very effective filter, and it is well-suited to preprocessing medical images where noise removal is required. While the VAE creates slightly blurry images, it has captured the underlying factors of generation for many anatomical sites, thereby demonstrating its usefulness for augmenting synthetic datasets in narrow-scope clinical datasets.